MCM / ICM • SECOND COLLECTION

Structures that explain decisions

12 composite figure studies / synthetic data / reproducible Python

N01 Allocation simplex and robustness

N02 Probability fields and search evidence

N03 Joint uncertainty with marginals

N04 Distribution ridges and threshold risk

N05 Constraint failure intersections

N06 Preference-driven rank changes

N07 Adaptive policy pathways

N08 Interaction chords and strongest links

N09 State probabilities and uncertainty

N10 Phase portrait and attraction basins

N11 Coalition attribution and ablation

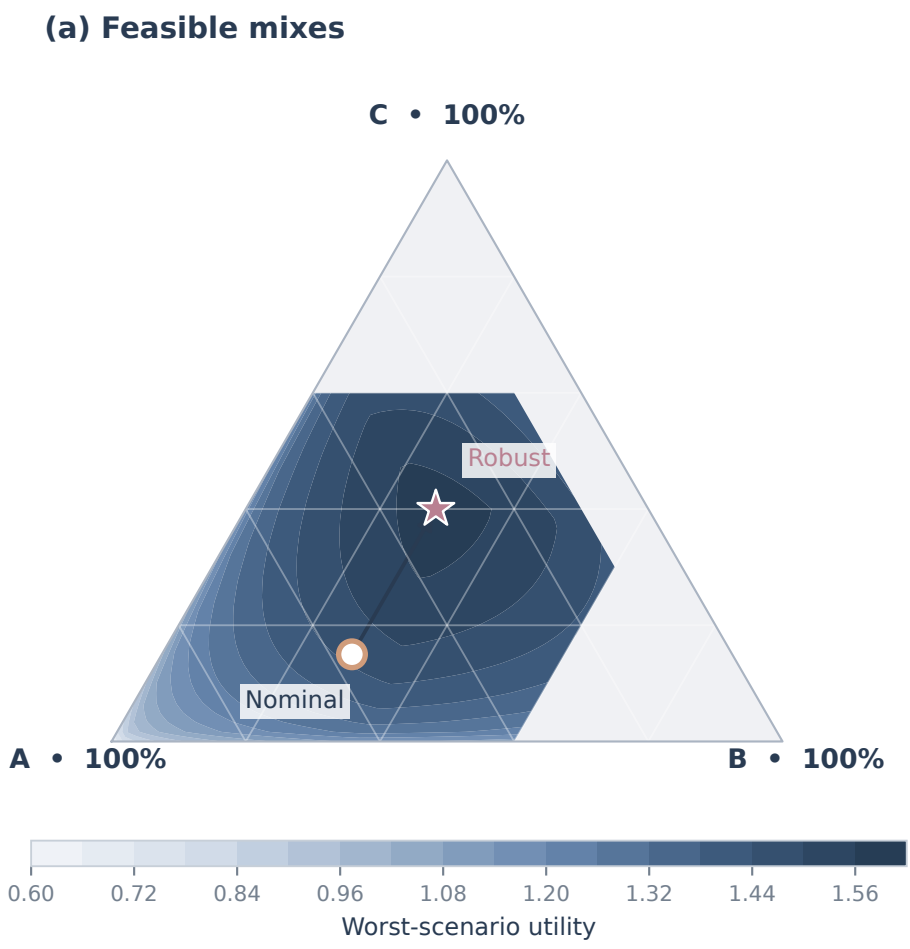
N12 Raincloud and extreme-loss behavior

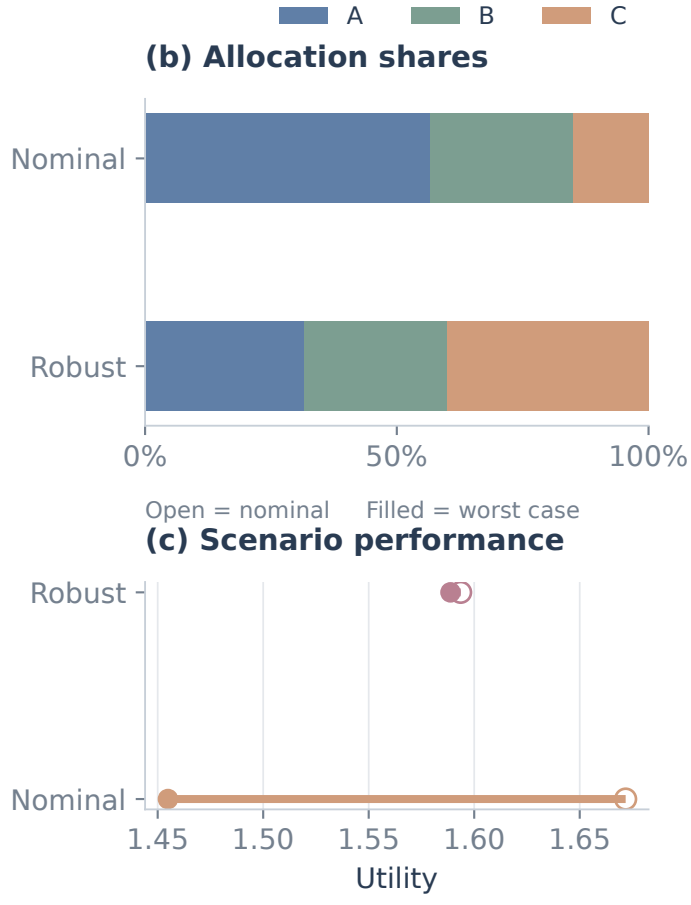
SLATE SAGE APRICOT IRIS DUSTY ROSE OLIVE

Sources and adaptation requirements are indexed in the accompanying Chinese guide.

Allocation geometry under uncertainty

A simplex exposes feasible mixes and the cost of protecting the worst scenario.

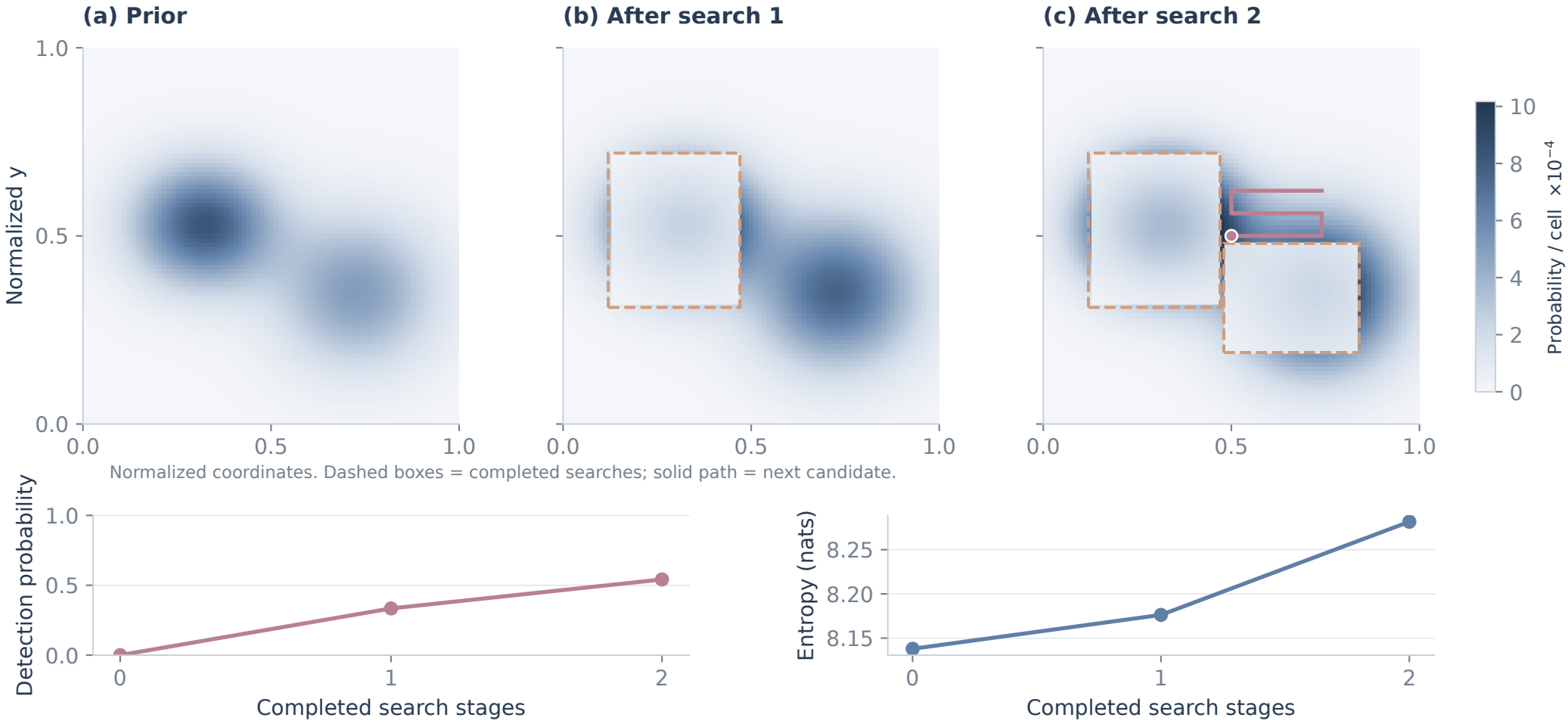




INPUT Three shares summing to 1 + scenario utilities + explicit constraints

Probability fields guide sequential search

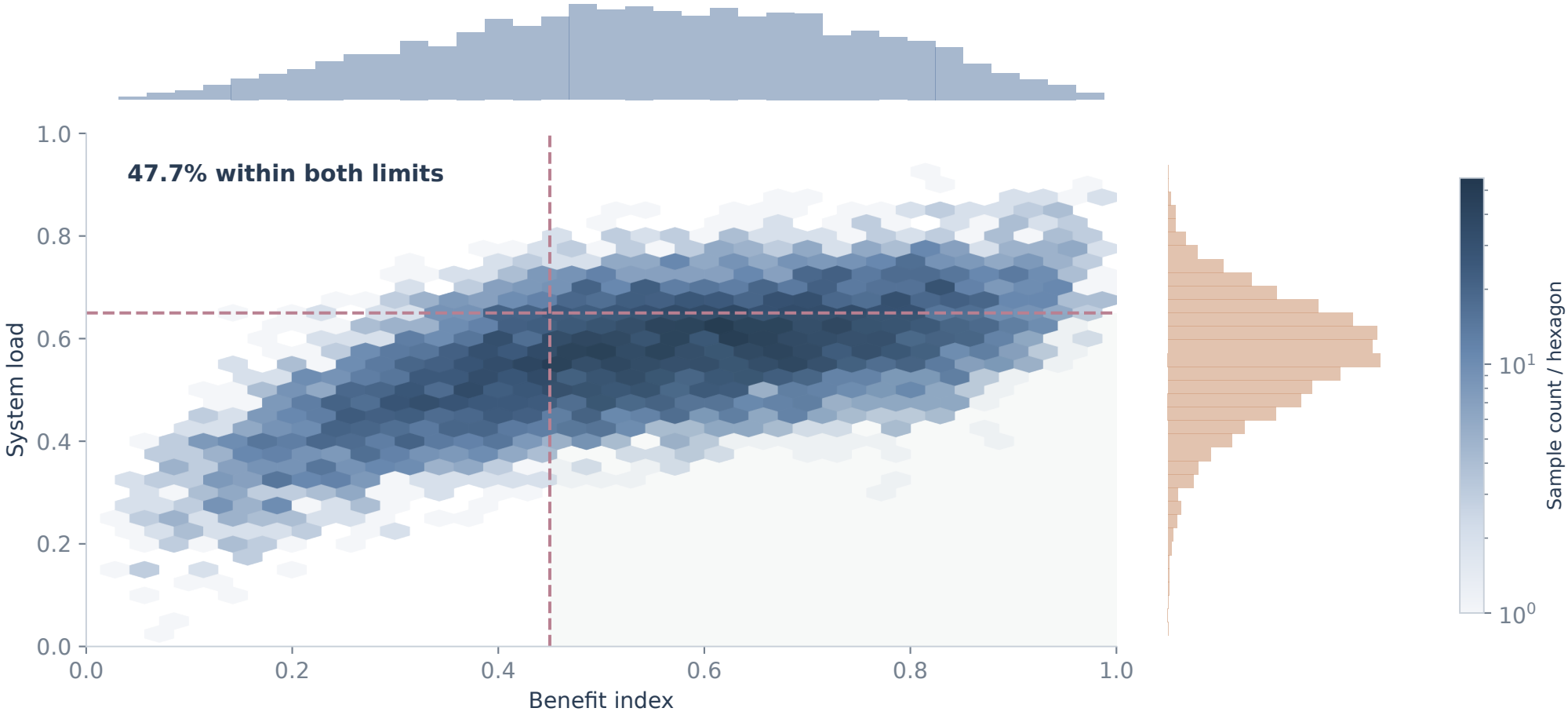
Negative evidence changes the distribution; coverage and uncertainty are read together.



INPUT Spatial prior + sensor likelihood + search footprints

Joint uncertainty and the operating window

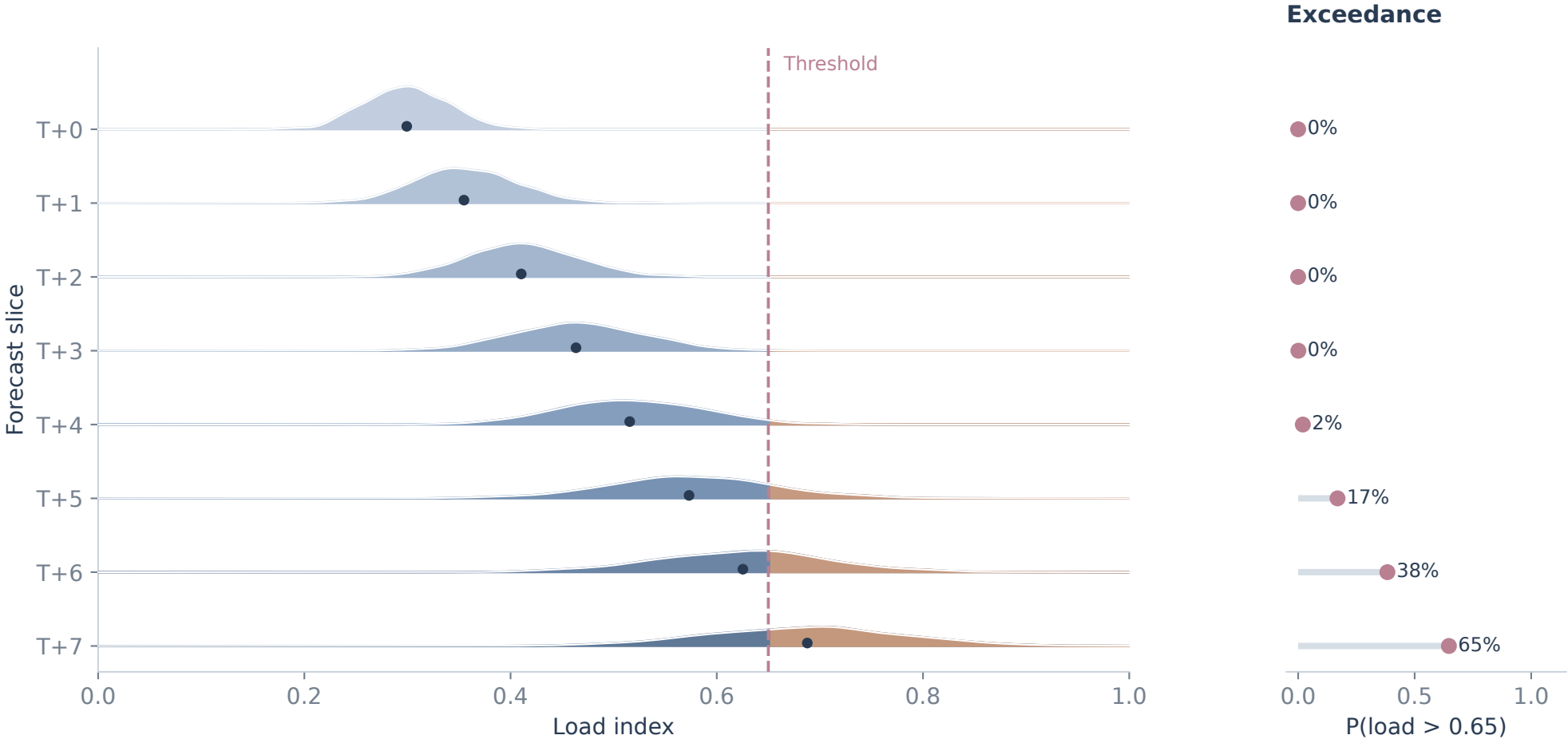
Marginals summarize each output; the center shows how the two outputs fail together.



INPUT Paired Monte Carlo outputs + two operational thresholds

Distributions approaching a critical threshold

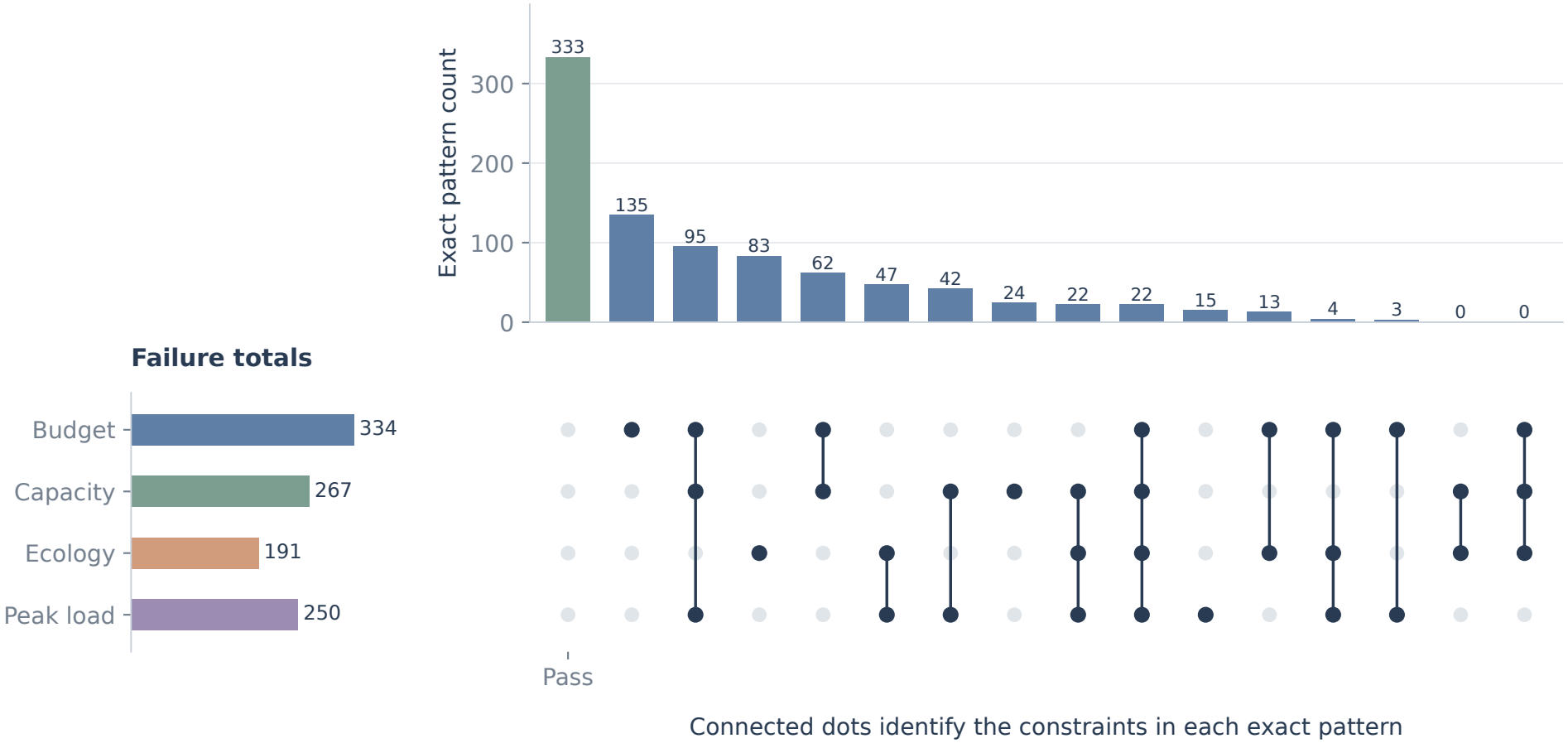
Ridgelines show shape; the aligned right panel quantifies exceedance risk.



INPUT Repeated output samples at each time slice + one threshold

Constraint failures that occur together

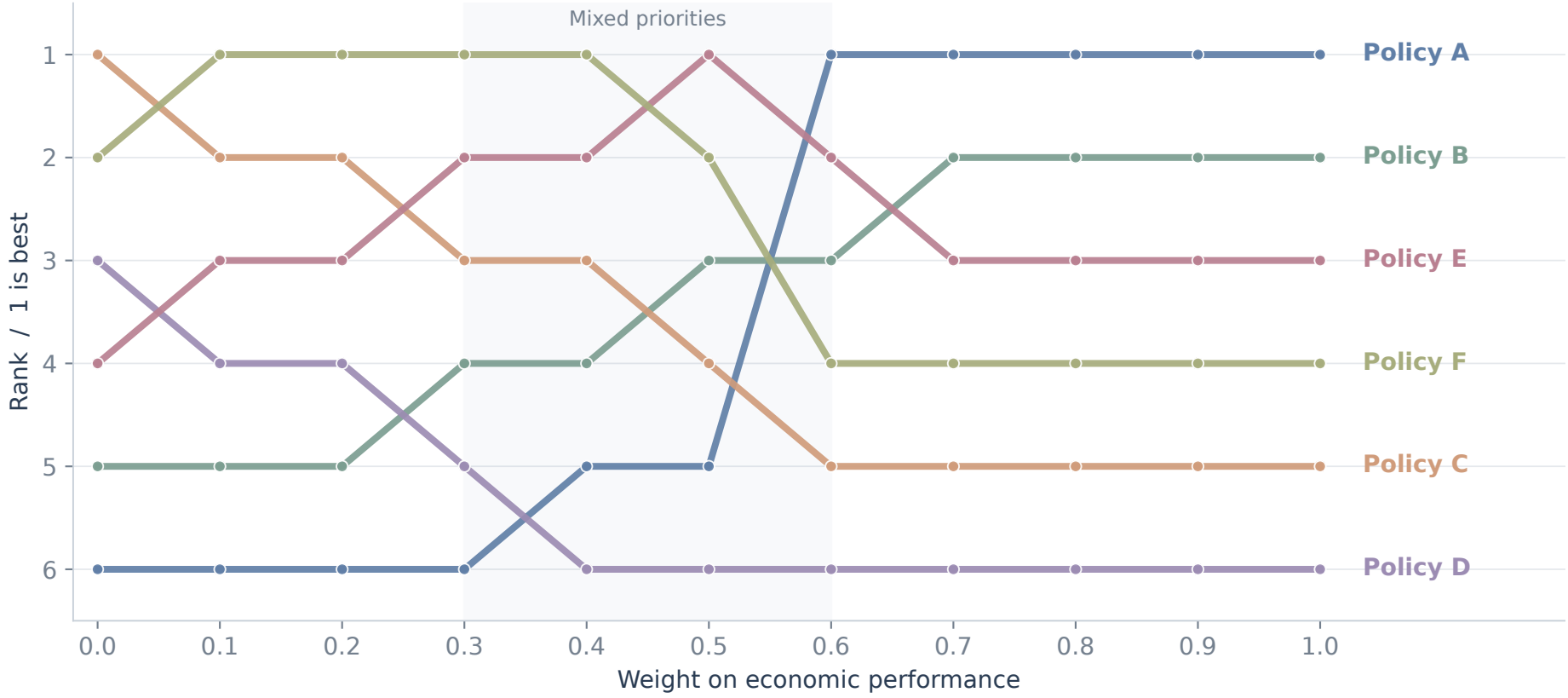
Exact intersections partition every stress case once; totals include overlapping failures.



INPUT One Boolean failure flag per constraint and per stress case

Policy rankings under changing priorities

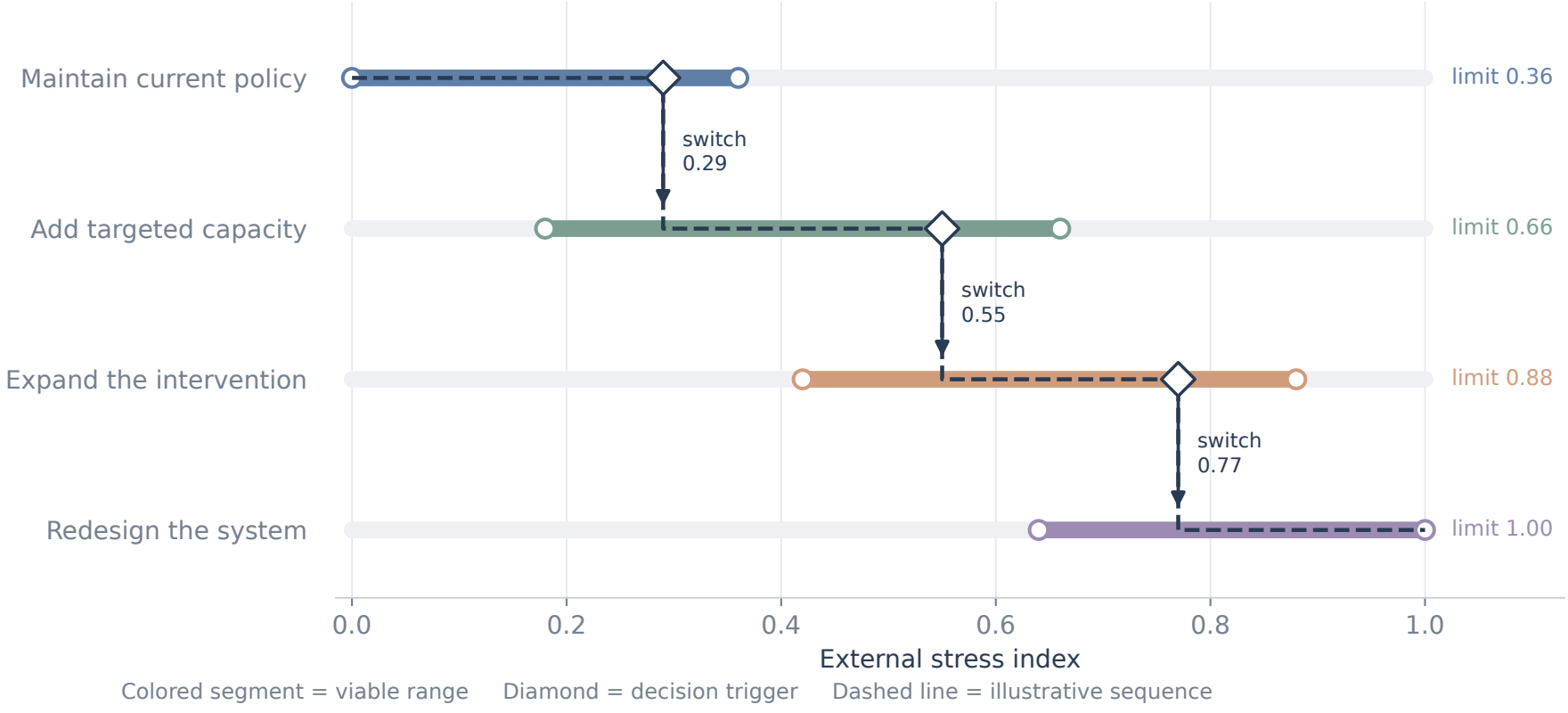
Lines connect eleven sampled weight settings; integer ranks are evaluated at the markers.



INPUT Policy-by-objective scores + sampled preference vectors

Adaptive actions as stress increases

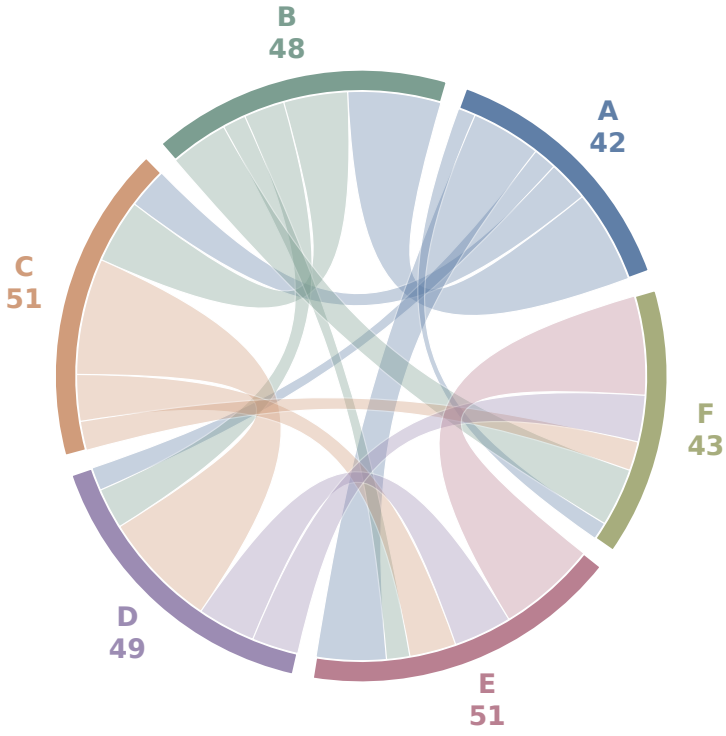
Availability windows, switching opportunities and one illustrative pathway share the same axis.



INPUT Action viability intervals + candidate switching thresholds

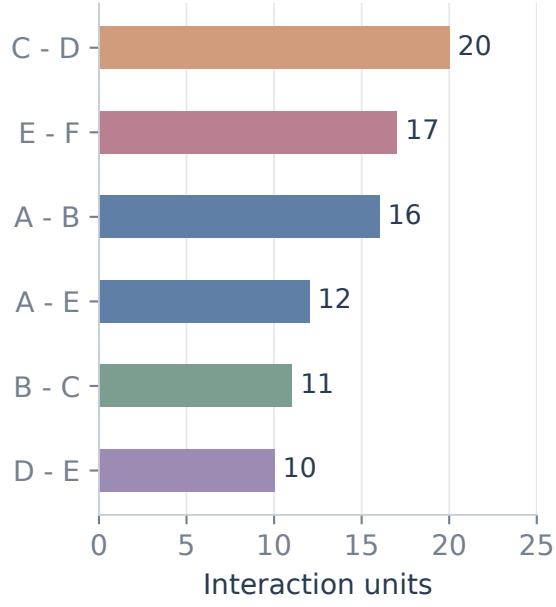
Exchanges between groups

Ribbon and sector widths encode the same symmetric interaction matrix.



142 total pair units

Strongest exchanges



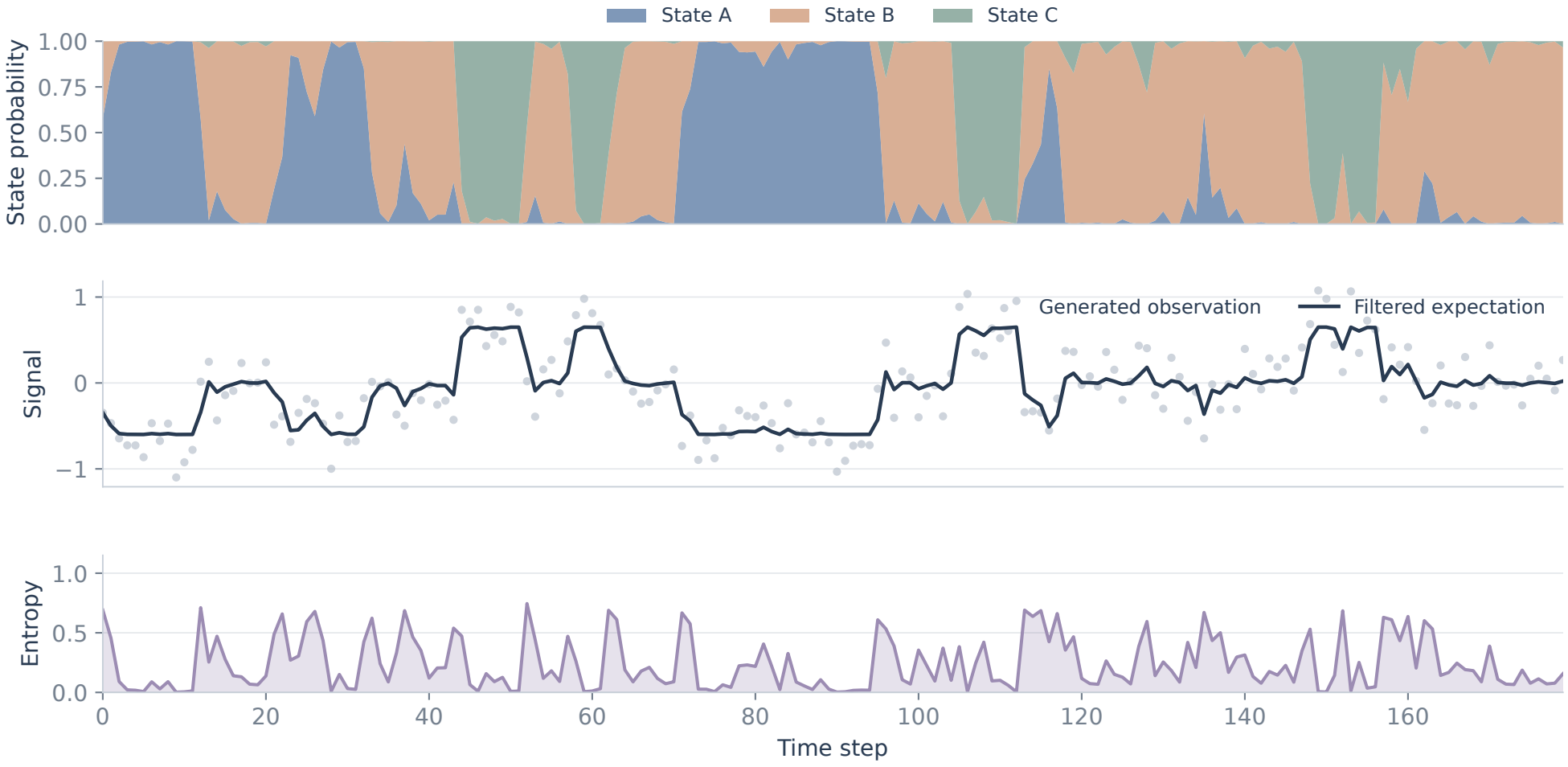
Undirected relationships

Node totals count incident flows

INPUT Small symmetric nonnegative interaction matrix

State changes under noisy observations

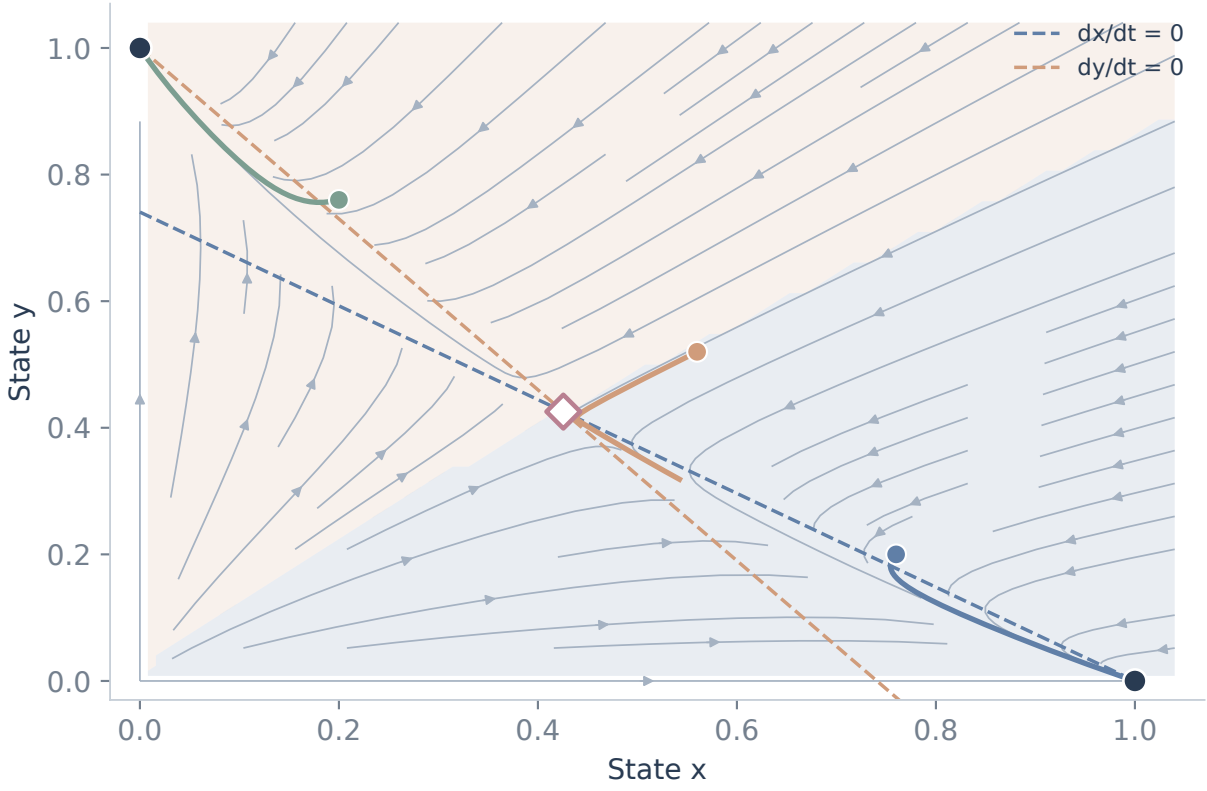
One shared timeline links observations, filtered state probabilities and posterior entropy.

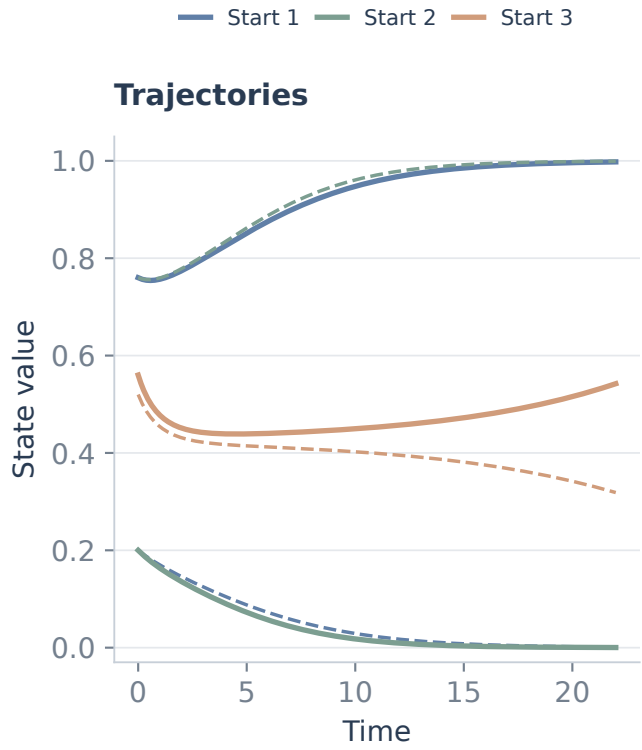


INPUT Ordered observations + state model + transition and observation probabilities

Initial conditions and long-run equilibria

Attraction basins, nullclines and time trajectories explain the same two-state ODE.





Solid = x Dashed = y

Filled point = stable equilibrium

Open diamond = saddle

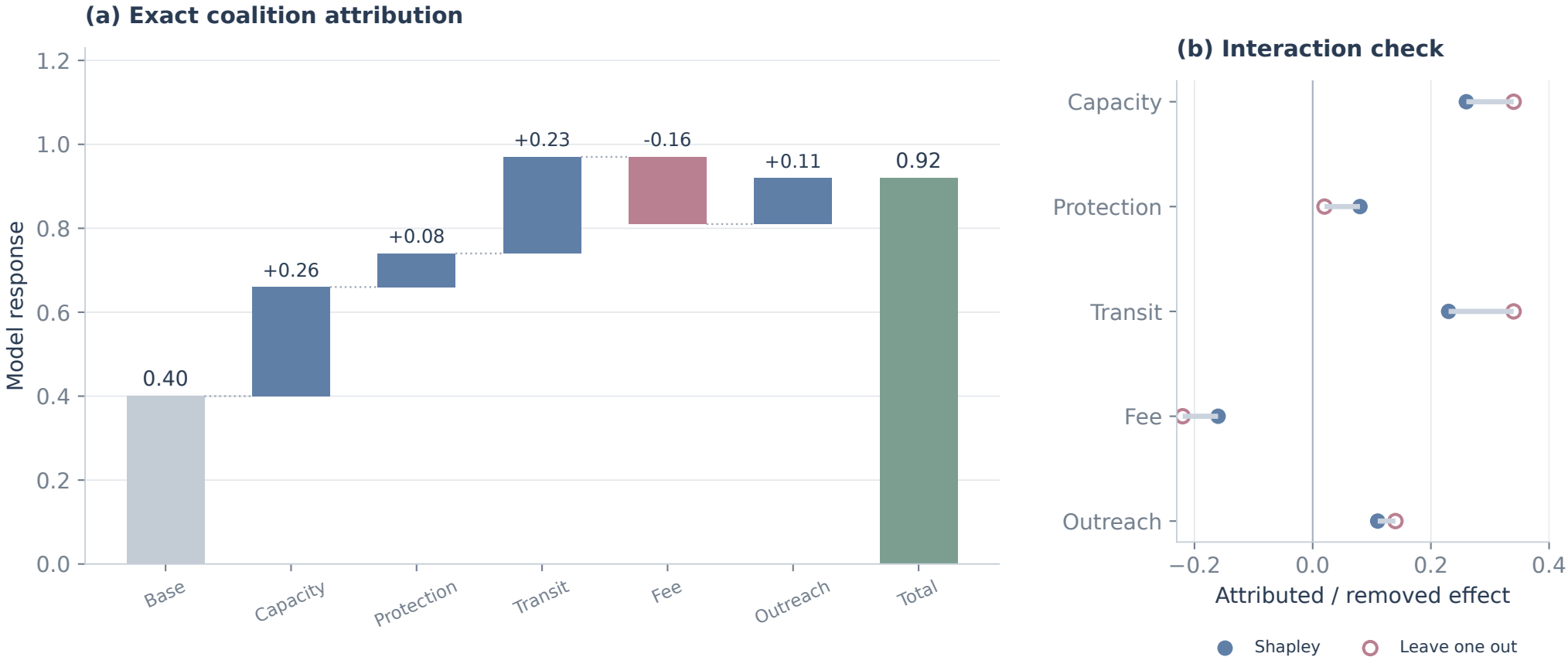
INPUT Two-state differential equations + initial states + integration horizon

SYNTHETIC DEMO / GENERATED VALUES AND SCHEMATIC COORDINATES

INSPIRATION P4 + own synthesis

Attribution and lever interactions

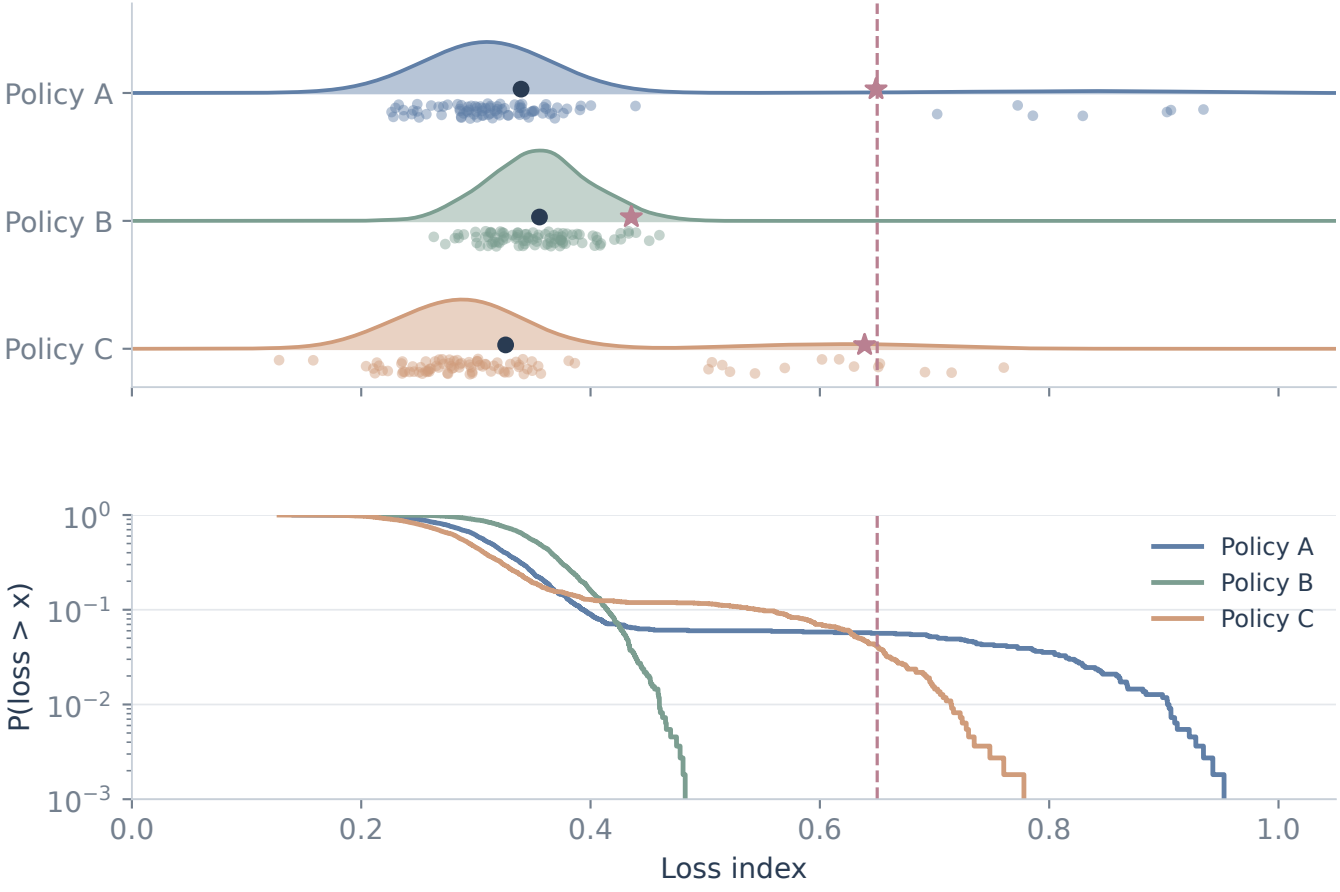
Coalition attribution shares interaction terms; leave-one-out effects retain the active context.



INPUT Model outputs for all 32 coalitions of five binary levers

Average performance and extreme-loss risk

Rainclouds retain individual samples; survival curves expose the high-loss tail.



Tail statistics

Policy	Mean	CVaR90
A	0.339	0.649
B	0.355	0.436
C	0.326	0.639

Dot = mean
Star = tail mean
CVaR90 = mean above sample 90th percentile

INPUT Repeated loss samples per policy + a tail probability level

Research and implementation references

P = inspected award paper / T = visualization source / layouts are original adaptations

Awards	COMAP 2024 official Outstanding Winner list
P1	2024 B 2407038 - probability-grid search, Figs. 8-10
P2	2024 B 2419984 - trajectories and search path, Figs. 6, 10, 12
P3	2024 C 2401298 - state, prediction and entropy, Figs. 4, 12-14
P4	2024 A 2400996 - disturbance and recovery, Figs. 10, 16
T1	mpltern - ternary geometry and contours
T2	UpSetPlot - exact intersections
T3	pyCirclize - chord geometry
T4	Allen et al. - Raincloud plots
T5	Matplotlib - hexbin and marginal distributions
T6	Crameri - scientific color-map design

Award status is checked against COMAP. Paper PDFs P1, P2 and P4 are public mirrored copies.

All sample values are generated. The Python renderer runs offline with NumPy, SciPy and Matplotlib.